



DIGITAL INVESTIGATION
— INTERNSHIP —

NETWORK RECONNAISSANCE AND SECURITY ASSESSMENT USING NMAP



PRACTICAL INTERNSHIP REPORT

A Practical Approach to Network Scanning,
Service Enumeration and Security Evaluation
Using Nmap in a Lab Environment

KEY HIGHLIGHTS



FOCUS

Network Reconnaissance
& Security Assessment



TOOLS USED

Kali Linux, Nmap, Windows 11,
VirtualBox, CMD



OBJECTIVE

Identify Active Hosts, Open Ports,
Running Services & Evaluate
Security Risks



ENVIRONMENT

Authorized Digital Investigation
Internship Lab



INTERNSHIP PERIOD

July 2026 – August 2026



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REPORT TYPE

Practical Internship Report



PURPOSE

Academic Learning &
Cybersecurity Skill Development



ETHICAL
RECONNAISSANCE



RESPONSIBLE
ASSESSMENT



STRONGER
DEFENSE



SECURE TODAY
SAFER TOMORROW

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the team at **Digital Investigation** for providing me with the opportunity to participate in this internship program and gain practical exposure to cyber investigation and network security concepts. The internship tasks and hands-on laboratory exercises provided through the [Digital Investigation Internship Portal](#) have significantly enhanced my understanding of cybersecurity fundamentals and professional investigation methodologies.

I am thankful to the mentors, coordinators, and technical support team for their continuous guidance, valuable feedback, and encouragement throughout this assignment. Their well-structured learning approach and practical exercises motivated me to perform every task systematically while following ethical cybersecurity practices.

I also acknowledge the use of my personal laboratory environment, consisting of a Windows host machine and a Kali Linux virtual machine, where all reconnaissance and security assessment activities were performed within the authorized scope defined by the internship guidelines. Conducting this assessment in a controlled environment strengthened my practical skills in network reconnaissance, service identification, security analysis, and technical documentation.

Finally, I express my gratitude to my college, faculty members, and everyone who supported and encouraged me throughout this learning journey. This practical assignment has increased my confidence in applying cybersecurity concepts to real-world scenarios and has motivated me to continue developing my technical and investigative skills.

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1. INTRODUCTION

In today's interconnected digital environment, organizations rely heavily on computer networks to support communication, business operations, cloud services, and data sharing. As network infrastructures continue to grow in complexity, they also become increasingly attractive targets for cyber threats such as unauthorized access, malware, ransomware, service exploitation, and data breaches. Identifying exposed services and understanding the overall security posture of a network are essential first steps in protecting critical systems and sensitive information.

Network reconnaissance is the initial phase of a cybersecurity assessment where information about target systems is collected in a controlled and authorized manner. This process involves identifying active hosts, discovering open ports, detecting running services, and understanding potential attack vectors. The information gathered during reconnaissance enables security professionals to evaluate network exposure, identify vulnerabilities, and recommend effective security controls before they can be exploited by attackers.

This report presents the findings of a practical **Network Reconnaissance and Basic Security Assessment** conducted within an authorized laboratory environment as part of the **Digital Investigation Internship Program**. The assessment was performed using **Kali Linux** and the **Nmap (Network Mapper)** utility to identify active hosts, analyze network services, and evaluate the security posture of a Windows-based system. All testing activities were conducted exclusively within a self-managed laboratory environment in accordance with ethical hacking principles and the internship guidelines.

The primary objective of this assessment is to demonstrate the practical application of network reconnaissance techniques, analyse the identified attack surface, assess potential security risks, and provide professional firewall and security hardening recommendations to improve the overall security posture of the target environment.

2. OBJECTIVES

The primary objectives of this practical network reconnaissance and security assessment are:

- To design a basic enterprise network topology consisting of Internet connectivity, a firewall, a DMZ web server, and two internal network segments.
- To perform host discovery using the Nmap utility and identify active systems within the authorized laboratory environment.
- To identify open TCP ports and detect running network services on the target Windows system.
- To analyse the collected reconnaissance data and identify potential attack surfaces that may expose the target system to security threats.
- To evaluate the security posture of the target environment based on the discovered services and network configuration.
- To recommend appropriate firewall rules and Access Control List (ACL) configurations to reduce unnecessary exposure and strengthen network security.
- To provide practical security hardening recommendations based on the findings obtained during the assessment.
- To perform all reconnaissance activities ethically within a self-managed laboratory environment while following the Digital Investigation Internship guidelines.

3. LAB ENVIRONMENT

The network reconnaissance and basic security assessment were conducted within a controlled and authorized laboratory environment. All scanning activities were performed exclusively on self-managed systems using industry-standard cybersecurity tools. The assessment complied with the Digital Investigation Internship guidelines and ensured that no unauthorized external systems or public networks were scanned.

Component	Configuration
Operating System (Host)	Microsoft Windows 11
Virtual Machine	Kali Linux (Latest Version)
Virtualization Platform	Oracle VirtualBox
Reconnaissance Tool	Nmap 7.98
Terminal Used	Kali Linux Terminal
Target System	Windows Host Machine
Target IP Address	192.168.1.24
Scanner IP Address	192.168.1.32
Network Type	Private Local Area Network (LAN)
Assessment Scope	Authorized Personal Laboratory

3.1 Lab Architecture

The assessment environment consisted of a Windows 11 host machine and a Kali Linux virtual machine connected within the same local network. Kali Linux was used as the attacking and reconnaissance platform, while the Windows host acted as the target system. Communication between both machines was verified using ICMP ping before initiating the reconnaissance phase with Nmap.

4. NETWORK TOPOLOGY DIAGRAM

The following network topology represents the simulated enterprise environment used during this assessment. The architecture consists of an Internet connection protected by a firewall, a DMZ containing a public web server, and an internal network where the Windows host machine and Kali Linux virtual machine are located. The reconnaissance activities were performed from the Kali Linux system against the authorized Windows host within the same local network.

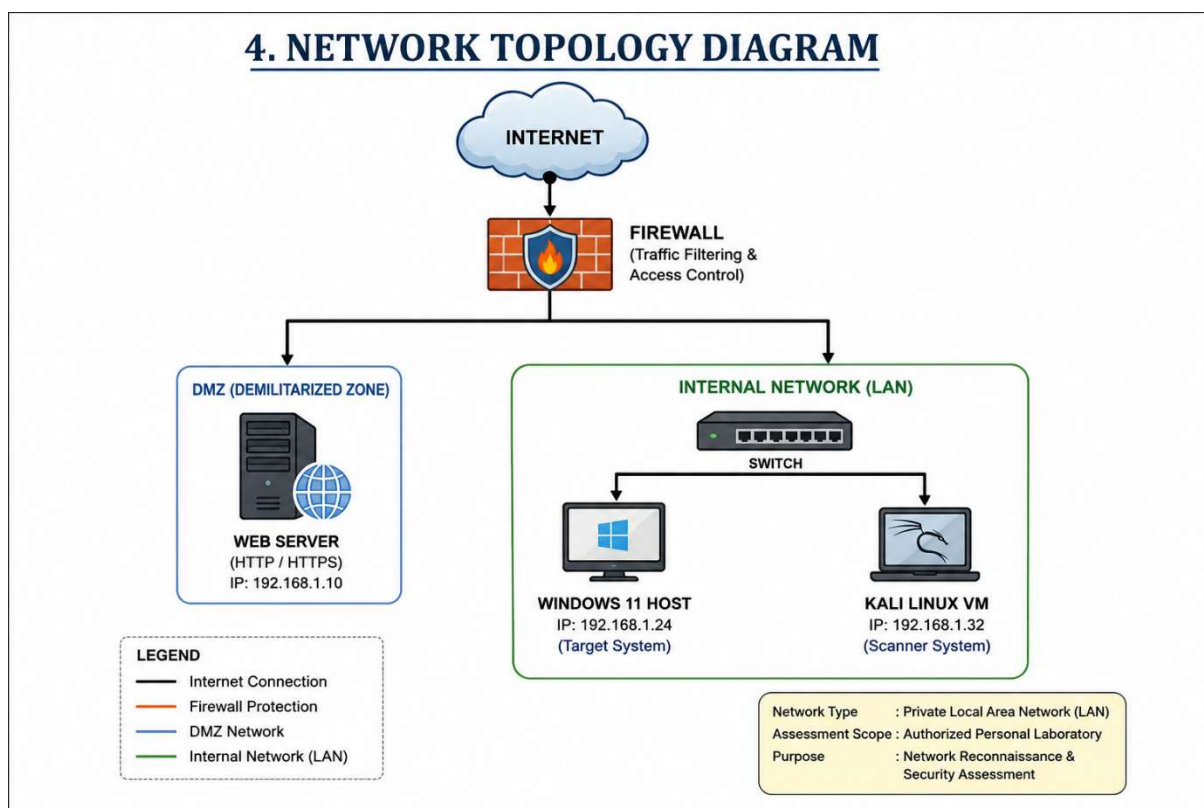


Figure 1: Enterprise Network Topology Used for the Security Assessment

5. TOOLS USED

The practical network reconnaissance and security assessment were conducted using industry-standard cybersecurity tools and software. Each tool served a specific purpose in identifying network hosts, detecting running services, collecting scan evidence, and documenting the overall security posture of the authorized laboratory environment.

Tool / Software	Version / Platform	Purpose
Kali Linux	Latest Version	Attack and reconnaissance platform
Microsoft Windows 11	Host Operating System	Target system
Oracle VirtualBox	VM Platform	Virtualization environment
Nmap	Version 7.98	Host discovery, port scanning, service detection
Windows Command Prompt (CMD)	Windows 11	Network information (IP configuration)
Kali Terminal	Linux Terminal	Executing reconnaissance commands
Microsoft Word	Microsoft 365	Report documentation
Draw.io / Custom Network Diagram	Online Tool	Network topology design

5.1 Tool Description

Kali Linux was used as the primary reconnaissance platform because it provides a wide collection of penetration testing and network security tools.

Nmap (Network Mapper) was used to discover active hosts, identify open TCP ports, detect running services, and collect evidence for the security assessment.

Oracle VirtualBox provided an isolated virtual laboratory where Kali Linux and the Windows host could communicate safely without scanning unauthorized systems.

6. NETWORK RECONNAISSANCE

Network reconnaissance is the process of gathering information about systems, devices, and services available within a network before performing a detailed security assessment. It is considered the first phase of both penetration testing and cyber investigations because it helps security professionals understand the target environment and identify potential attack surfaces.

During this assessment, reconnaissance activities were performed only within an authorized personal laboratory environment in accordance with the Digital Investigation Internship guidelines. The assessment focused on identifying active hosts, detecting open TCP ports, discovering running services, and collecting technical evidence from the target Windows system using the Nmap utility.

Before initiating the port scanning process, network connectivity between the Kali Linux virtual machine and the Windows host system was verified using ICMP ping. Once successful communication was confirmed, multiple Nmap scan techniques were executed to gather information about the target system while maintaining an ethical and controlled assessment approach.

6.1 Assessment Workflow

Step	Activity Performed
1	Verified network connectivity using ICMP Ping
2	Identified active host using Nmap Host Discovery
3	Performed Service Version Detection
4	Executed Aggressive Scan (-A)
5	Collected scan evidence and screenshots
6	Analysed open ports and security risks
7	Prepared firewall and hardening recommendations

7. NMAP COMMANDS USED

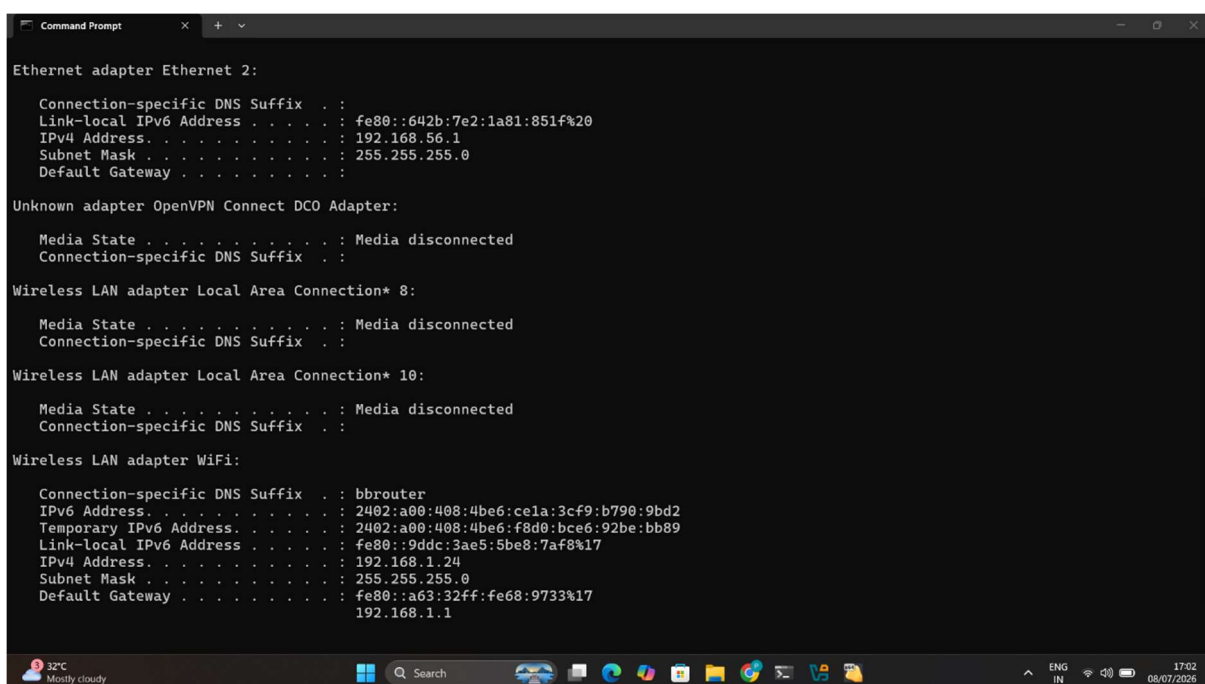
Before performing the reconnaissance activities, the IP addresses of both the Windows host machine and the Kali Linux virtual machine were identified to verify that both systems were connected to the same local network. After confirming connectivity, ICMP ping was used to test communication, followed by multiple Nmap scanning techniques to identify active hosts, detect running services, and collect technical evidence from the target system.

7.1 Windows Host IP Configuration

The Windows Command Prompt was used to obtain the IPv4 configuration of the target system. The IP address identified during the assessment was **192.168.1.24**, which served as the target for the reconnaissance activities.

Command used:

Ipconfig

A screenshot of a Windows Command Prompt window titled "Command Prompt". The window displays the output of the 'ipconfig' command. It shows details for three network adapters: Ethernet adapter Ethernet 2, Unknown adapter OpenVPN Connect DCO Adapter, and Wireless LAN adapter Local Area Connection* 8. The Ethernet adapter shows an IPv4 address of 192.168.56.1. The OpenVPN adapter shows a media state of 'Media disconnected'. The Local Area Connection* 8 adapter also shows a media state of 'Media disconnected'. The Wireless LAN adapter Local Area Connection* 10 shows an IPv4 address of 192.168.1.24. The Wireless LAN adapter WiFi shows an IPv4 address of 192.168.1.1. The taskbar at the bottom shows the system clock as 17:02 on 08/07/2026.

```
Ethernet adapter Ethernet 2:
    Connection-specific DNS Suffix . . : 
    Link-Local IPv6 Address . . . . . : fe80::642b:7e2:1a81:851f%20
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Unknown adapter OpenVPN Connect DCO Adapter:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . . : 

Wireless LAN adapter Local Area Connection* 8:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . . : 

Wireless LAN adapter Local Area Connection* 10:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . . : 

Wireless LAN adapter WiFi:
    Connection-specific DNS Suffix . . : bbrouter
    IPv6 Address. . . . . : 2402:a00:408:4be6:ce1a:3cf9:b790:9bd2
    Temporary IPv6 Address. . . . . : 2402:a00:408:4be6:f8d0:bce6:92be:bb89
    Link-local IPv6 Address . . . . . : fe80::9ddc:3ae5:5be8:7af8%17
    IPv4 Address. . . . . : 192.168.1.24
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::a63:32ff:fe68:9733%17
                                192.168.1.1
```

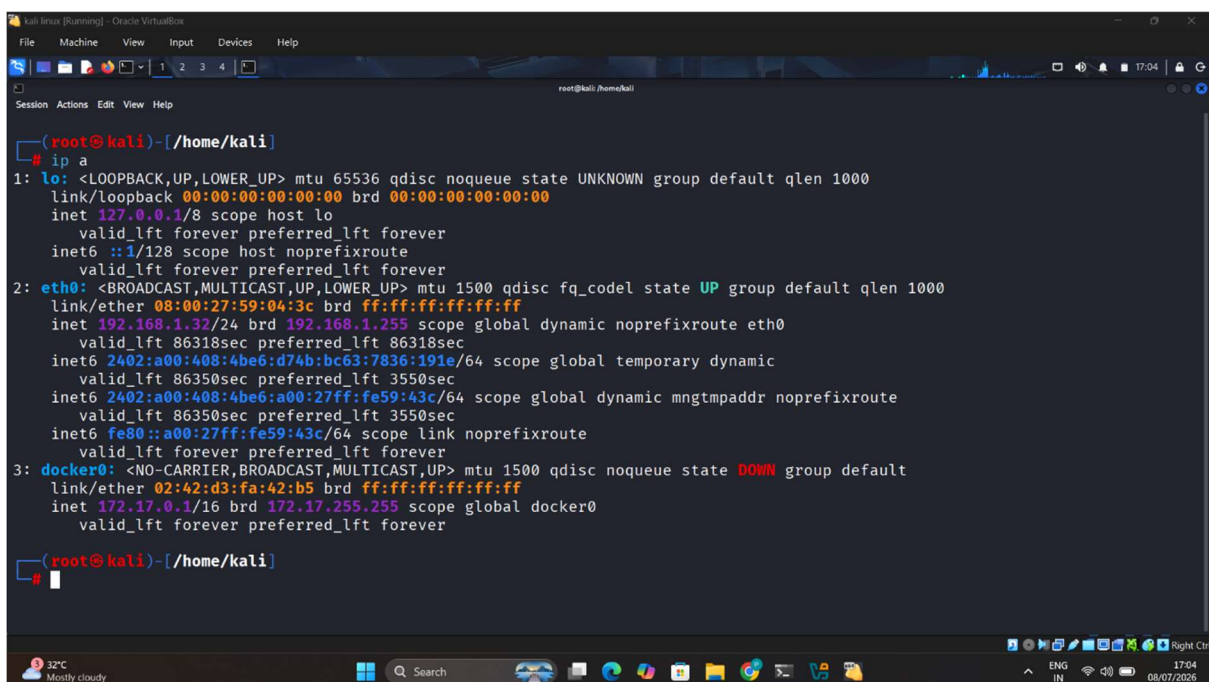
Figure 2: Windows Host IP Configuration using ipconfig

7.2 Kali Linux Network Configuration

The Kali Linux virtual machine was configured within the same local network and assigned the IP address **192.168.1.32**. This system was used to perform all reconnaissance and scanning activities.

Command used:

`ip a`



```
(root@kali)-[/home/kali]
# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:59:04:3c brd ff:ff:ff:ff:ff:ff
   inet 192.168.1.32/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
       valid_lft 86318sec preferred_lft 86318sec
   inet6 2402:a00:408:4be6:d74b:bc63:7836:191e/64 scope global temporary dynamic
       valid_lft 86350sec preferred_lft 3550sec
   inet6 2402:a00:408:4be6:a00:27ff:fe59:43c/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86350sec preferred_lft 3550sec
   inet6 fe80::a00:27ff:fe59:43c/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether 02:42:d3:fa:42:b5 brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever

(root@kali)-[/home/kali]
```

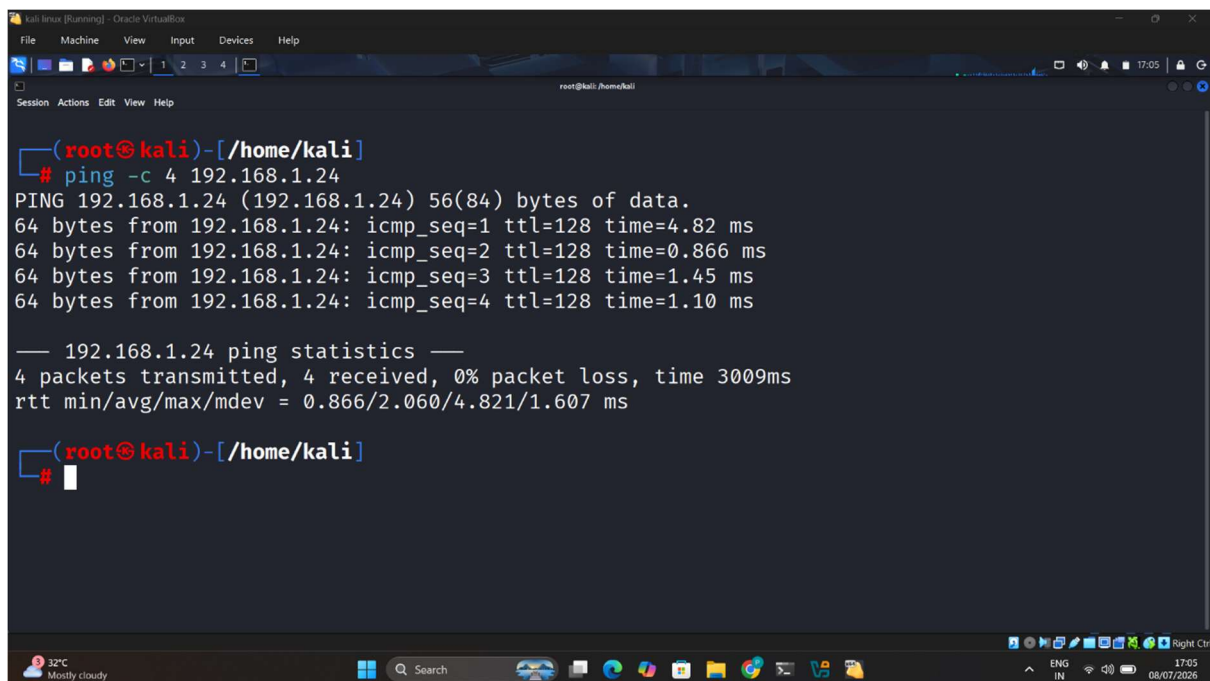
Figure 3: Kali Linux Network Configuration using ip a

7.3 Connectivity Verification

Before initiating the Nmap scan, network connectivity between the Kali Linux virtual machine and the Windows host was verified using ICMP Ping. Successful replies confirmed that both systems were reachable within the same local network.

Command used:

```
ping -c 4 192.168.1.24
```



```
(root@kali)-[/home/kali]
# ping -c 4 192.168.1.24
PING 192.168.1.24 (192.168.1.24) 56(84) bytes of data.
64 bytes from 192.168.1.24: icmp_seq=1 ttl=128 time=4.82 ms
64 bytes from 192.168.1.24: icmp_seq=2 ttl=128 time=0.866 ms
64 bytes from 192.168.1.24: icmp_seq=3 ttl=128 time=1.45 ms
64 bytes from 192.168.1.24: icmp_seq=4 ttl=128 time=1.10 ms

— 192.168.1.24 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 0.866/2.060/4.821/1.607 ms

(root@kali)-[/home/kali]
#
```

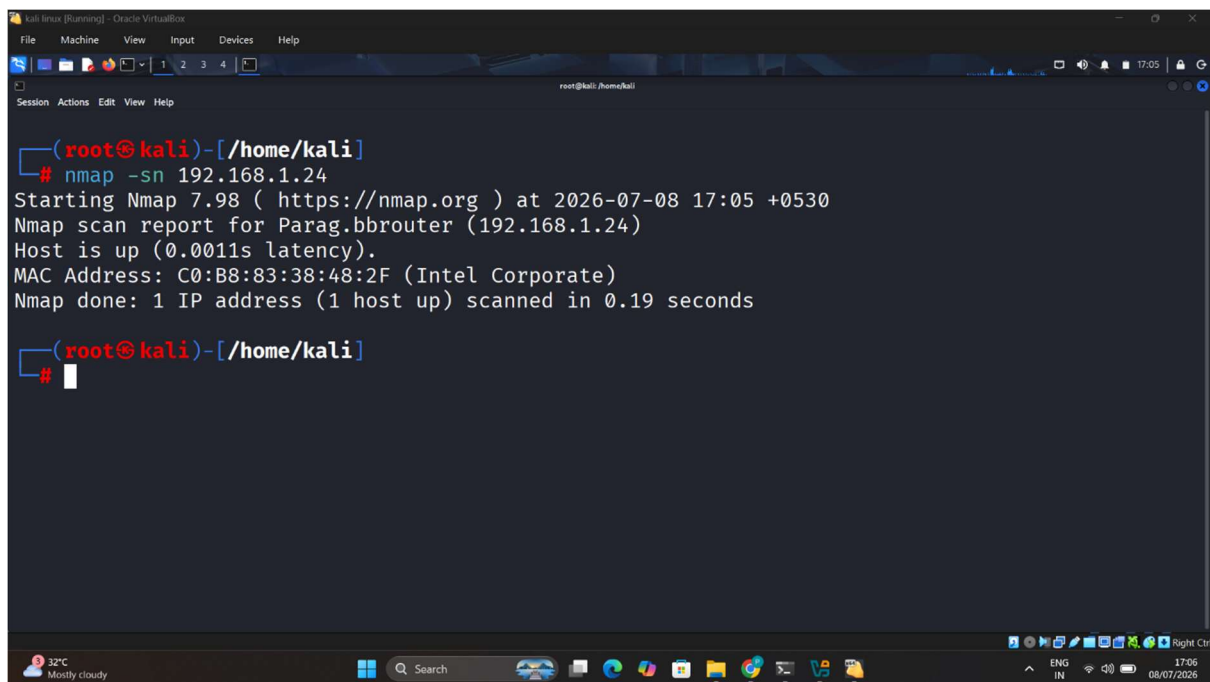
Figure 4: ICMP Connectivity Verification between Kali Linux and Windows Host

7.4 Host Discovery Scan

After confirming network connectivity, a host discovery scan was performed using the Nmap -sn option. This scan identifies whether the target system is online without performing a full port scan. The objective of this phase was to verify that the Windows host was reachable before proceeding with service detection and advanced reconnaissance.

Command Executed

```
nmap -sn 192.168.1.24
```

A screenshot of a Kali Linux terminal window running inside an Oracle VM VirtualBox. The terminal shows the command 'nmap -sn 192.168.1.24' being executed. The output indicates that the host is up, with a latency of 0.0011s, and provides the MAC address C0:B8:83:38:48:2F (Intel Corporate). The scan was completed in 0.19 seconds. The terminal window has a dark background with red and blue text for the prompt and command. The VirtualBox window title is 'kali linux (Running) - Oracle VM VirtualBox'. The bottom of the image shows the Windows taskbar with various icons and the system clock displaying 17:05 on 08/07/2026.

```
(root@kali)-[/home/kali]
# nmap -sn 192.168.1.24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-08 17:05 +0530
Nmap scan report for Parag.bbrouter (192.168.1.24)
Host is up (0.0011s latency).
MAC Address: C0:B8:83:38:48:2F (Intel Corporate)
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds

(root@kali)-[/home/kali]
#
```

Figure 5: Host Discovery Scan Using Nmap (-sn)

Observation

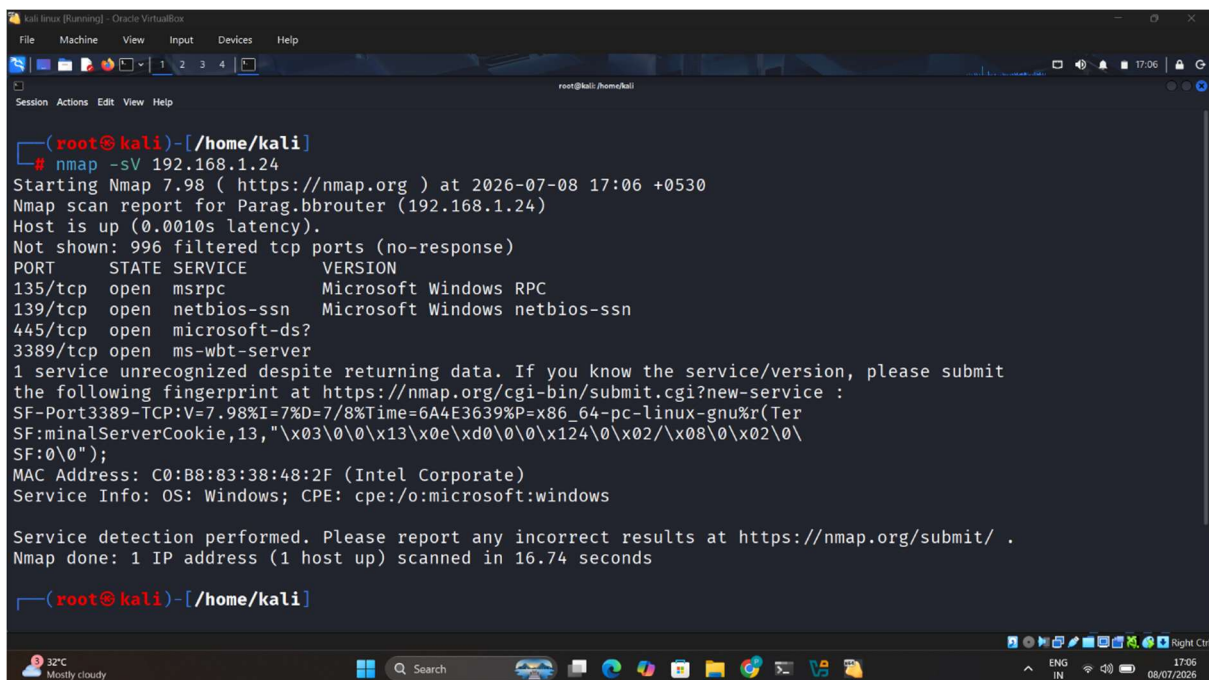
The host discovery scan successfully identified the target Windows system as active on the network. The scan confirmed that the target IP address (192.168.1.24) was reachable and available for further reconnaissance activities. No additional hosts were included in the assessment, as the scope of the internship was limited to authorized personal systems only.

7.5 SERVICE VERSION DETECTION

After verifying that the target system was online, a Service Version Detection scan was performed using the Nmap -sV option. This scan identifies open TCP ports and attempts to determine the services running on those ports. Service identification enables security analysts to understand which applications are exposed to the network and evaluate potential security risks associated with each service.

Command Executed

```
nmap -sV 192.168.1.24
```



```
(root@kali)-[/home/kali]
# nmap -sV 192.168.1.24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-08 17:06 +0530
Nmap scan report for Parag.bbrouther (192.168.1.24)
Host is up (0.0010s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3389/tcp   open  ms-wbt-server
1 service unrecognized despite returning data. If you know the service/version, please submit
the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port3389-TCP:V=7.98I=7%D=7/8%Time=6A4E3639P=x86_64-pc-linux-gnu%r(Ter
SF:minalServerCookie,13,"\\x03\\0\\0\\x13\\x0e\\xd0\\0\\0\\x124\\0\\x02\\x08\\0\\x02\\0\\
SF:0\\0");
MAC Address: C0:B8:83:38:48:2F (Intel Corporate)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.74 seconds

(root@kali)-[/home/kali]
```

Figure 6: Service Version Detection Using Nmap (-sV)

Scan Findings

Port	State	Detected Service	Security Impact
135/tcp	Open	Microsoft RPC	Required by Windows but may reveal system information.

139/tcp	Open	NetBIOS Session Service	Legacy protocol that should be restricted on modern networks.
445/tcp	Open	Microsoft SMB	High-value target for attackers if not properly secured.
3389/tcp	Open	Microsoft Remote Desktop (RDP)	Remote access service that should not be exposed publicly.

Observation

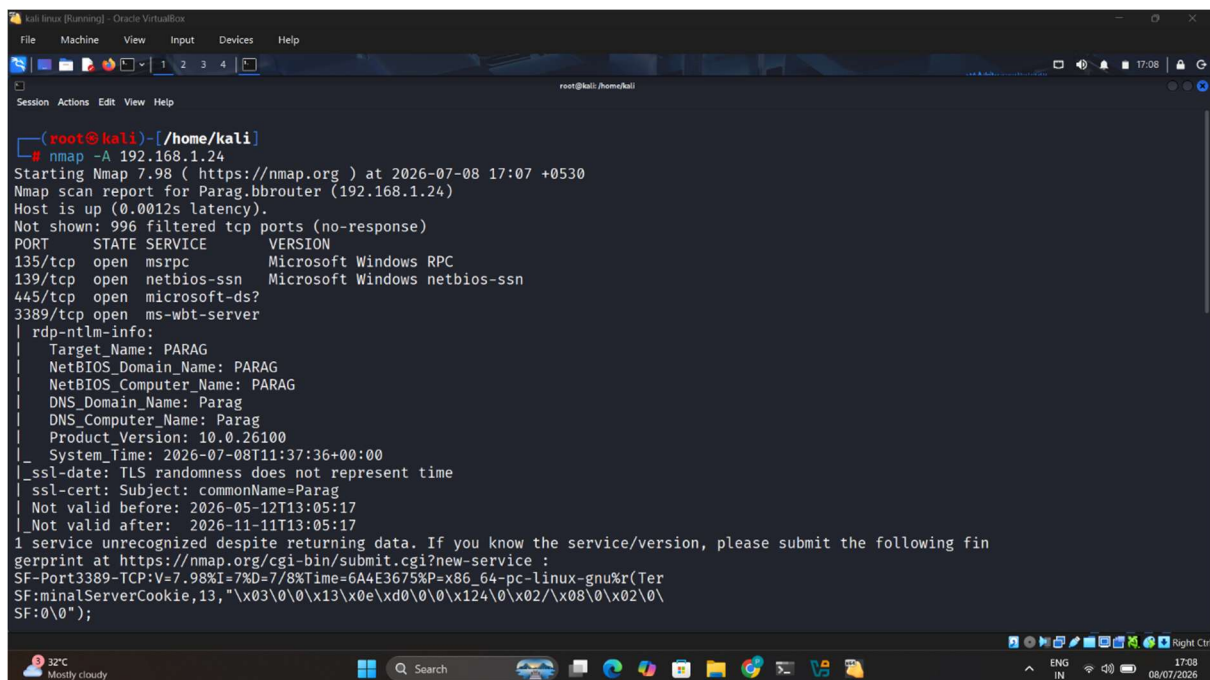
The Service Version Detection scan successfully identified multiple Windows services running on the target machine. The presence of RPC, NetBIOS, SMB, and Remote Desktop services indicates that the system provides standard Windows networking functionality. However, these services also increase the attack surface if they are not properly protected through firewall rules, strong authentication, regular updates, and network segmentation.

7.6 AGGRESSIVE SCAN

To obtain detailed information about the target system, an aggressive scan was performed using the Nmap -A option. This scan combines multiple advanced reconnaissance techniques, including operating system detection, service version detection, default script scanning, and traceroute analysis. The purpose of this scan was to collect comprehensive information about the target system while operating within the authorized laboratory environment.

Command Executed

```
nmap -A 192.168.1.24
```



```
(root@kali)~[/home/kali]
# nmap -A 192.168.1.24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-08 17:07 +0530
Nmap scan report for Parag.bbrouter (192.168.1.24)
Host is up (0.0012s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3389/tcp   open  ms-wbt-server
rdp-ntlm-info:
|_ Target_Name: PARAG
|_ NetBIOS_Domain_Name: PARAG
|_ NetBIOS_Computer_Name: PARAG
|_ DNS_Domain_Name: Parag
|_ DNS_Computer_Name: Parag
|_ Product_Version: 10.0.26100
|_ System_Time: 2026-07-08T11:37:36+00:00
|_ ssl-date: TLS randomness does not represent time
|_ ssl-cert: Subject: commonName=Parag
|_ Not valid before: 2026-05-12T13:05:17
|_ Not valid after: 2026-11-11T13:05:17
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port3389-TCP:V=7.98%I=7%D=7/8%Time=6A4E3675%P=x86_64-pc-linux-gnu%r(Ter
SF:minaIServerCookie,13,"x03\0\0\x13\x0e\xd0\0\0\x124\0\x02/\x08\0\x02\0\
SF:\0\0");
```

Figure 7: Aggressive Scan Using Nmap (-A)

Information Gathered

Information	Observation
Target Host	Windows 11 Host
Target IP Address	192.168.1.24

Scanner Machine	Kali Linux (192.168.1.32)
Open Ports	135, 139, 445, 3389
Running Services	Microsoft RPC, NetBIOS, SMB, Remote Desktop
Scan Type	Aggressive Scan (-A)
Scan Status	Successfully Completed

Observation

The aggressive scan provided a comprehensive overview of the target system by combining multiple reconnaissance techniques into a single operation. The scan confirmed the availability of several Windows services and validated the results obtained from previous scans. This information forms the basis for identifying security risks, evaluating the attack surface, and recommending appropriate firewall rules and security hardening measures.

8. SECURITY ASSESSMENT

The security assessment was performed based on the reconnaissance data collected during the Nmap scanning process. The objective of this assessment was to identify exposed network services, evaluate the potential attack surface, and determine possible security risks associated with the target Windows host. The findings presented below are based solely on the authorized laboratory environment used during this internship assignment.

8.1 Open Ports Identified

Port	Service	Purpose	Risk Level
135/TCP	Microsoft RPC	Remote Procedure Call communication	Medium
139/TCP	NetBIOS Session Service	Legacy Windows file sharing	Medium
445/TCP	Microsoft SMB	File and Printer Sharing	High
3389/TCP	Remote Desktop Protocol (RDP)	Remote Administration	High

8.2 Running Services Analysis

The Nmap service detection scan identified multiple Windows networking services running on the target host. These services are commonly enabled in Windows environments to support system management, file sharing, and remote administration. Although they are legitimate services, improper configuration or unnecessary exposure may increase the overall attack surface of the system. Security controls such as firewall filtering, strong authentication, regular patching, and network segmentation should be implemented to reduce associated risks.

8.3 Security Risk Summary

Finding	Risk	Recommendation
RPC Service Exposed	Medium	Restrict unnecessary remote access using Windows Firewall.
NetBIOS Enabled	Medium	Disable NetBIOS if it is not required.
SMB Service Available	High	Disable SMBv1, apply regular security updates, and restrict access.
RDP Enabled	High	Restrict RDP access to trusted IP addresses and enable Multi-Factor Authentication (MFA).

Assessment Summary

The assessment identified several Windows services that are commonly found in enterprise environments. While these services are required for normal system operation, they should not be unnecessarily exposed to untrusted networks. The use of proper firewall rules, secure authentication mechanisms, system updates, and continuous monitoring can significantly reduce the likelihood of unauthorized access and network-based attacks.

9. ATTACK SURFACE ANALYSIS

The attack surface of a system represents all possible entry points that an attacker may attempt to exploit. During this assessment, the Nmap scans identified several active Windows services exposed on the target host. Although these services are essential for normal system functionality, improper configuration, weak authentication, or outdated software may increase the likelihood of unauthorized access. A detailed analysis of each exposed service is presented below.

9.1 Attack Surface Analysis

Open Port	Potential Threat	Possible Impact
135 (RPC)	Information Gathering	Attackers may identify Windows services and collect system information.
139 (NetBIOS)	Enumeration	Hostname, shared resources, and network details may be disclosed.
445 (SMB)	SMB Exploitation	Unauthorized access, malware spread, ransomware, and lateral movement if vulnerabilities exist.
3389 (RDP)	Brute Force Attack	Unauthorized remote login if weak passwords or poor access controls are used.

9.2 Overall Risk Evaluation

Based on the scan results, the target system exposes several Windows networking services commonly found in enterprise environments. Since the assessment was performed within a private laboratory network, the immediate risk is limited. However, if similar services were directly accessible from the Internet without proper firewall protection, they could significantly increase the attack surface and become attractive targets for cyber attackers. Appropriate firewall policies, strong authentication, timely software updates, and continuous monitoring are essential to minimize these risks.

9.3 Key Findings

- The Windows host was successfully identified and responded to reconnaissance requests.
- Four TCP ports were found to be open during the assessment.
- Multiple Windows networking services were detected using Nmap.
- The SMB and Remote Desktop services represent the most significant attack surface if exposed outside the trusted network.
- The assessment confirmed that the reconnaissance activities were successfully completed within the authorized laboratory environment.

10. FIREWALL & ACL RECOMMENDATIONS

Based on the reconnaissance findings and the services identified during the Nmap assessment, several firewall and access control recommendations are proposed to improve the security posture of the target environment. These recommendations aim to minimize unnecessary exposure, reduce the attack surface, and protect the system from unauthorized access while maintaining legitimate network functionality.

10.1 Recommended Firewall Rules

Rule No.	Firewall / ACL Rule	Purpose
1	Allow HTTP (TCP 80) only to the DMZ Web Server.	Protect internal systems while allowing public web access.
2	Allow HTTPS (TCP 443) only to the DMZ Web Server.	Ensure secure encrypted communication for web services.
3	Block SMB (TCP 445) from untrusted networks.	Prevent unauthorized file-sharing access and malware propagation.
4	Block NetBIOS (TCP 139) where not required.	Reduce information disclosure and legacy protocol exposure.
5	Restrict Remote Desktop (TCP 3389) to authorized administrator IP addresses only.	Prevent unauthorized remote access attempts.
6	Deny all inbound traffic by default unless explicitly permitted.	Follow the Principle of Least Privilege.
7	Allow ICMP (Ping) only for internal troubleshooting when required.	Reduce unnecessary network visibility.

8	Enable firewall logging and monitor denied connections.	Improve security monitoring and incident investigation.
9	Separate the DMZ from the internal network using firewall policies.	Prevent attackers from moving into internal systems if the web server is compromised.
10	Review and update firewall rules periodically.	Maintain an effective and up-to-date security posture.

10.2 Benefits of Firewall Implementation

- Reduces the overall attack surface.
- Blocks unauthorized inbound network connections.
- Protects sensitive internal resources.
- Prevents unauthorized remote administration.
- Limits the spread of malware across the network.
- Improves visibility through firewall logging and monitoring.
- Supports compliance with cybersecurity best practices.
- Enhances the overall security posture of the organization.

Recommendation Summary

Implementing these firewall and access control recommendations will significantly improve the security of the assessed environment. Restricting unnecessary services, enforcing least-privilege access, enabling logging, and regularly reviewing firewall policies are essential practices for protecting enterprise networks against modern cyber threats.

11. SECURITY HARDENING RECOMMENDATIONS

Security hardening is the process of reducing the attack surface of a system by implementing protective security controls and eliminating unnecessary risks. Based on the findings of the network reconnaissance and security assessment, the following recommendations are proposed to improve the security posture of the Windows host and reduce the possibility of unauthorized access or exploitation.

11.1 Recommended Security Hardening Measures

Recommendation	Security Benefit
Keep Windows fully updated with the latest security patches.	Reduces the risk of known vulnerabilities being exploited.
Disable SMBv1 and other legacy protocols if not required.	Prevents attacks targeting outdated network services.
Use strong passwords and enable Multi-Factor Authentication (MFA).	Protects against brute-force and credential theft attacks.
Restrict Remote Desktop (RDP) access to trusted administrator systems only.	Minimizes unauthorized remote access attempts.
Enable Windows Defender Firewall and review firewall rules regularly.	Blocks unnecessary inbound and outbound connections.
Disable unused services and unnecessary open ports.	Reduces the overall attack surface.
Install and maintain updated antivirus or endpoint protection software.	Detects and blocks malware and malicious activities.
Perform regular system backups and verify recovery procedures.	Ensures data can be restored after hardware failure or ransomware attacks.

Enable system logging and security event monitoring.	Improves incident detection and forensic investigation capabilities.
Conduct periodic vulnerability assessments using authorized tools.	Helps identify and remediate security weaknesses before they are exploited.

11.2 Security Best Practices

- Follow the Principle of Least Privilege (PoLP).
- Regularly review user accounts and access permissions.
- Remove unused applications and services.
- Monitor security logs for suspicious activities.
- Perform periodic network security assessments.
- Keep cybersecurity tools and operating systems up to date.
- Educate users about phishing, malware, and password security.
- Test backup and disaster recovery procedures regularly.

Summary

Implementing these security hardening measures will significantly strengthen the security posture of the assessed environment. Regular updates, secure configuration, continuous monitoring, and proactive risk management are essential practices for protecting modern enterprise systems against evolving cyber threats.

12. CONCLUSION

This practical assessment successfully demonstrated the fundamental concepts of network reconnaissance and basic security assessment within an authorized laboratory environment. Using the Nmap utility, active hosts were identified, open ports were discovered and running services on the target Windows system were successfully analysed. The collected evidence provided valuable insight into the network exposure and overall security posture of the target environment.

The assessment identified multiple Windows services, including Microsoft RPC, NetBIOS, SMB, and Remote Desktop Protocol (RDP). Although these services are commonly required for normal system operation, they also increase the potential attack surface if not properly secured. Appropriate firewall configurations, strong authentication, timely software updates, and continuous monitoring are essential to reduce associated security risks.

This internship task provided practical experience in performing ethical network reconnaissance, analysing security findings, and preparing a professional cybersecurity assessment report. The knowledge and skills gained through this activity have strengthened my understanding of network security, vulnerability assessment, and cyber investigation methodologies. The experience will also serve as a valuable foundation for future cybersecurity projects and professional responsibilities.

13. REFERENCES

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